

Adithya V. Sastry

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🌐 <https://avs-origami.github.io>

PERSONAL PROFILE

I am an undergraduate student studying Electrical and Computer Engineering at Duke University. My primary interest lies in exploring how co-designing novel microprocessor architectures and software paradigms can enable the creation of small-scale, power-efficient, yet robust robotic systems. My research experience includes reinforcement learning, robotic control theory, low-power electronics, and energy-harvesting systems, and I additionally have experience with PCB design, rapid prototyping with microcontrollers and various sensors, and coding in various programming languages. Further, I am a quick and eager learner, and am very good at learning unfamiliar concepts on-the-fly – it's one of the main reasons I enjoy engaging in research.

ACADEMICS **GPA: 4.7** **SAT: 1580** **ACT: 35**

- AP Calculus BC (5)
- Linear Algebra
- Differential Equations
- AP Physics C (5)
- AP Chemistry (5)
- AP CSA (5)

SKILLS

- Programming: Rust, C++, Python, Java, Bash
- Circuit + PCB Design: Autodesk Eagle, Fritzing
- Machine Learning: Torch (Python and C++), JAX, PPO
- Tools: Linux, Git, LaTeX

AWARDS AND RECOGNITIONS

Research Science Institute Scholar (2025)

- One of 62 US students (100 internationally) selected to participate in a 6-week research internship at MIT

Southern Appalachian Science & Engineering Fair (2019 – 2025)

- Overall Senior Division Fourth Place (2024)
- ORNL Computer Science Award (2024)
- Overall Junior Division Fifth Place (2022)
- 2X Overall Junior Division Grand Prize (2021 & 2019)
- Overall Junior Division Third Place (2020)
- 2X Junior Division Winner for Outstanding Projects in Engineering, Tau Beta Pi (2022 & 2020)
- 3X Lemelson Early Inventor Prize (2022, 2021, 2020)
- 3X Winner for Best Demonstrations of Electrical or Electronics Design or Computer Science, IEEE (2025, 2022, 2020)

RESEARCH INTERNSHIPS

REALM Lab at Massachusetts Institute of Technology

June 2025 – July 2026, average 14 hours per week, peak 35 hours per week during RSI 2025 program.

Mentor – Prof. Chuchu Fan, MIT AeroAstro.

Fluidic City Lab at University of Tennessee, Knoxville

April 2024 – September 2026, average 20 hours per week, peak 35 hours per week. Mentor – Prof. Weizi Li, EECS.

Authored paper currently under review at IROS 2026 workshops; [preprint available on arXiv](#).

EPIC Lab at University of Tennessee, Knoxville

May 2023 – September 2023; 35 hours per week.

Mentor – Prof. Sai Swaminathan, EECS.

Contributed to [peer-reviewed research paper published in ACM Journal](#), IMWUT, Sept 2024.

PROJECTS

Safe Robotic Control for Autonomous Aircraft Swarms (2025-2026)

Worked on extending an algorithm for safe reinforcement learning control of multi-agent robotic systems beyond the initially tested environments to a robotic system with more complex control dynamics. Presented this work at the 2025 Research Science Institute Symposium.

Learning-Guided Sparsification of Dynamic Graphs in Robotic Exploration (2023-2026)

Developed an algorithm enabling autonomous robotic exploration of unknown environments. Extended under mentorship of Prof. Weizi Li, exploring use of reinforcement learning to prune RRT used by exploration algorithm to carry out pathfinding and frontier detection. Demonstrated method of pruning exploration graph which generalizes to long-horizon exploration tasks in highly complex environments, improving exploration efficiency by up to 2.1 percentage points and boosting runtimes by up to 2x. [Preprint on arXiv](#) and [source code on GitHub](#).

Uninformed Search Technique Based Autonomous Exploration Robot (2022)

Designed and built a robot integrating a microcomputer, motors, sensors, and a custom-designed PCB to enable fully autonomous navigation in random environments. Programmed the microcomputer in MicroPython to process sensor data, track robot's location, and

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AWARDS AND RECOGNITIONS

Southern Appalachian Science & Engineering Fair (2019 – 2025)

- 3X Excellence in Science and Engineering, Office of Naval Research (2022, 2021, 2020)
- Numerous other special awards (Ricoh Americas, AIChE, ASM Materials and Education Foundation, JTV Brilliance Award, Knox County Superintendent's Award, Qardian Labs Award for Excellence in Problem Solving)

BROADCOM Masters Competition

- Selected in the Top 300 MASTERS among 3,476 entrants throughout the US, Project: “Material Identification Using Arduino Powered Surface Spectroscopy for Improved Recycling”, 2020

3M Young Scientist Challenge

- Tennessee State Merit Winner in 2020 for “RecycleSense – An Arduino Powered Surface Spectroscopic Device for Improved Household Recycling”
- Tennessee State Merit Winner in 2019 for “DriveAlive – A Microcontroller Based Distracted Driving Monitor”

Davidson Institute for Talent Development

- Davidson Young Scholar, 2016 – Present. Qualified for the Young Scholar Program in 3rd grade based on individually administered achievement testing and comprehensive application process.

Other Competitions

- International Research Olympiad 2025 Semifinalist
- Research and Preparation award at AltMUN 2024
- Second Place in Pre-Calculus, Tennessee Math Teachers Association Math Contest, 2023
- Third Place in Algebra II, Tennessee Math Teachers Association Math Contest, 2022

OTHER INTERESTS

- Created Tin Can Linux, my own custom Linux distribution, and continue to maintain it
- South Indian Classical (Carnatic) vocal performance
- Piano, Origami, Bonsai, Chess, Tennis, Debating

PROJECTS

identify previously visited areas. This robot shows potential as a tool for search-and-rescue missions and exploration of hazardous or inaccessible environments.

CO₂ Capture from Vehicle Exhaust Using the Dry Carbonate Process (2021)

Investigated CO₂ capture from vehicle exhaust using sodium carbonate, building on methods used in power plants. Presented the project at the 2021 Southern Appalachian Science & Engineering Fair, winning the Grand Prize in the Junior Division and multiple category awards. Also presented this project at the 2023 Tennessee Junior Science and Humanities Symposium.

Custom Arduino and PCB Design (2021)

Designed a custom, quarter-sized Arduino variant powered by a LiPo battery for hobby projects. Developed PCB schematics using Autodesk Eagle and, after fabrication, assembled the board using oven reflow and hand soldering techniques. This unique Arduino variant is not available on the market. I used it actively in various personal projects to validate its functionality.

Material Identification Using Arduino Powered Surface Spectroscopy for Improved Household Recycling (2020)

Built a device using an Arduino and a spectrometer breakout board to identify common household recyclable materials, aiming to enhance the quality of single-stream recycling. Presented the project at the 2020 Southern Appalachian Science & Engineering Fair, the 2020 3M Young Scientist Challenge, and the 2020 Broadcom Masters competition, earning multiple awards.

VOLUNTEERING

UpTech Project (2024 – Current)

Founded a new initiative to upcycle laptops and provide them to graduating seniors who cannot afford a laptop for college; chapters established in three schools

AIM for Seva Non-Profit Org (2022 – Current)

Coordinating volunteering activities; Fundraising for child sponsorships; Training local children to give vocal performances at fundraising events

Origami USA Knoxville Chapter (2019 – 2020)

Organizing monthly meetings and special events